



POLITECNICO
MILANO 1863

MPI

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Rules

- Rename the `ants_simulation_XX.c` file replacing `XX` with the number of your group
- Write in the comment on top of the file your group number and the name of all group members
- Submit only a single c file with your solution
 - Submitted from the contact email provided in the group registration document

Requirements

- You are to implement a simple simulator of the behavior of a colony of ants
- Ants move in a one-dimensional space x
 - The minimum value of x is 0 and the maximum is 1000
- In the space, there are 10 sources of food
 - Their positions are stored in the `food_sources` array
- We define the center of the colony as the average value of the positions of all ants in the colony

Requirements

- The simulation evolves in discrete rounds for a given number of iterations
 - `num_iterations` in the template file
- Before the first iteration
 - Process P0 computes the positions of all ants and distributes them to the other processes
 - Use the `init_ants` function to obtain the positions
 - Process P0 computes the positions of food sources and distributes them to the other processes
 - Use the `init_food_sources` function to obtain the positions

Requirements

- At each iteration, each ant moves according to two forces $F1$ and $F2$
 - $F1$ attracts the ant towards the nearest source of food
 - If $d1$ is the distance between the ant and the nearest source of food, the strength of $F1$ is $0.01*d1$
 - $F2$ attracts the ant towards the center of the colony
 - If $d2$ is the distance between the ant and the center of the colony, the strength of $F2$ is $0.012*d2$
- The position of the ant at the end of the iteration will be the old position + $F1$ + $F2$
 - Notice that $F1$ and $F2$ may point to opposite directions!

Requirements

- At each iteration, you have to compute and print the center of the colony
 - The code for printing is already in the template
- Run the simulation in parallel on multiple processes, minimizing synchronization and communication as much as possible